PHANIDHAR AKULA

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Simulation Systems, Digital Twins & Geospatial Computing • Authorized to work in the U.S. Oct 2026 – Oct 2029 (OPT + STEM OPT)

EDUCATION

Miami University | M.S. in Computer Science *Aug 2024 – Aug 2026*
Oxford, OH • GPA: 3.86/4.00 • Full Scholarship • Assistantship
Thesis: SimForge – A Reproducible, Cross-Simulator Benchmarking Framework for Urban Traffic Simulation (committee of three faculty advisors)
Coursework: Generative AI, Machine Learning, Advanced Database Systems, Software Quality Assurance, Cryptography

Malla Reddy University | B.Tech in Computer Science *Aug 2020 – May 2024*
Hyderabad, India • CGPA: 8.2/10 • Two peer-reviewed publications (CNN soil classification, LDA topic modeling)

SELECTED PROJECTS

SimForge | Master's Thesis – A Reproducible, Cross-Simulator Benchmarking Framework for Urban Traffic Simulation *Aug 2024 – Present*
Python, OpenMP, multiprocessing, SLURM, OSC HPC (Pitzer / Cardinal / Ascend), SUMO, MATSim, DTALite, OSM, US Census PUMS

- Integrated three heterogeneous traffic engines (SUMO micro+meso, MATSim queue-mobsim, DTALite DTA) under one canonical five-file scenario bundle (network, demand, signals, config, manifest with SHA-256 hashes); byte-identical reproducibility guarantees across re-runs with code-enforced fair-comparison contract.
- Designed a state-aware BFS heuristic pre-router with OSM turn-restriction enforcement and strongly connected component (SCC) feasibility filtering, forcing engines onto byte-identical canonical routes so any cross-engine travel-time delta is provably an engine-internal property, not an input asymmetry.
- Compressed BFS pre-routing bottleneck from ~68h to ~5–8h (~25–30× speedup) on NYC 500K-trip scenarios via canonical-route deduplication, 16-way multiprocessing.Pool parallelism, and a JSONL content-addressed cache; deployed across three OSC HPC clusters with cluster-specific SBATCH tuning.
- Scaled to ~80K nodes / ~200K directed links across Chicago, NYC, LA at five demand tiers (1K → 500K trips), calibrated against US Census PUMS microdata (2.4M–6.8M persons) via ModelGen, lifting realism from ~20–40% baseline to ~70–72%. Validated by ~570 pytest tests, mutation tests on load-bearing modules, byte-identity guards, and Student's-t 95% CIs; 76% coverage.

Cityscape | Open-Source Population Synthesis & Geospatial Demand Generation *Aug 2025 – Present*
C++, OpenMP, OSM / Geofabrik, US Census PUMS, OSC HPC (Pitzer)

- Authored three merged C++ pull requests (github.com/raodj/cityscape, PRs #1–#3) to an open-source PUMS-driven population synthesizer and trip-demand generator. Fixed OSM building-classification and orphan-ring logic across five metros, collapsing synthetic-home ratios (e.g. LA ~70% → ~5%) and removing ~1.25M stranded-population artifacts, affecting well over 1M building records.
- Exposed search-radius CLI flags and ran a 4-city, 16-combination HPC parameter sweep on OSC Pitzer, raising worst-case calibration R^2 from 0.812 to 0.917. Contributions integrated into the SimForge thesis demand pipeline.

LumiAI | Production AI Education Platform – studywithlumi.com *Nov 2025 – Present*
React, TypeScript, Supabase (PostgreSQL/Auth/Storage), Anthropic Claude, Vercel serverless, custom domain

- Solo-architected, built, deployed, and operate a production AI tutoring platform serving 55+ active student users; document-grounded AI chat, auto-generated quizzes and flashcards, SM-2 spaced-repetition review, and voice mode. Owned the full stack with row-level security, server-side API-key isolation, Google OAuth, and an admin analytics panel.

EXPERIENCE

Miami University | Graduate Assistant – Department of CSE *Aug 2025 – May 2026*
Oxford, OH • Research group of Dr. DJ Rao

- Graduate research in the simulation / digital-twin group; lead developer on Cityscape (above), with work integrated into the SimForge thesis pipeline. Co-administer the Grand Challenges Scholars Program (GCSP) mentoring undergraduate research.
- Earlier industry experience (2023–2024): software engineering intern at Smart Pathfinders Overseas Education and freelance full-stack engineer for three early-stage startup clients, building React / TypeScript and Python applications with REST APIs, MySQL, and AWS.

LEADERSHIP

- President, Graduate Students of Color Association (104 members; Certificate of Appreciation, 2026) and elected Board Member, Graduate Student Council, Miami University (Aug 2025 – May 2026) – led recruitment, events, cross-team coordination, and reviewing student petitions.

TECHNICAL SKILLS

Languages: Python (advanced), C++, JavaScript, TypeScript, SQL, Java (basic)
AI/ML & Data: PyTorch, TensorFlow, CNNs, LDA / topic modeling, LLM API integration, REST APIs, GraphQL, MySQL, PostgreSQL, AWS
HPC & Systems: OpenMP, multiprocessing, SLURM, OSC clusters (Pitzer / Cardinal / Ascend), Linux, Docker, Git, content-addressed caching
Engineering & Evaluation: benchmarking & evaluation frameworks, statistical validation (95% CIs), byte-identity reproducibility guards, pytest, mutation testing, code review
Simulation & Geospatial: SUMO, MATSim, DTALite, agent-based & traffic simulation, digital twins, OSM / Geofabrik, US Census PUMS pipelines, network modeling

PUBLICATIONS & RECOGNITION

- Akula, P., Stojanovski, S., & Rao, D. M. “CITYSCAPE: A Digital Twin Framework for Temporospatial Modeling of Work-Commute Dynamics and Roadway Infrastructure Demand” Frontiers (first author; under review, 2026).
- Two peer-reviewed publications in IJSREM (2023): *"Topic Modelling of Web Pages with LDA Methods"* (DOI: 10.55041/IJSREM27350) · *"Soil Image Classification using Deep Learning"* (DOI: 10.55041/IJSREM23138).
- 10+ technical certifications: AWS Cloud Foundations, Applied ML & AI, Software Engineering (AICTE, Coursera, NPTEL, Forage, AWS).